DECO2300 Interactive Prototype 1 - TikTok XR Dance Studio

Testing plan for interactive prototype 1

This project is a recreation of TikTok in XR to let users learn and film TikTok dances of themselves as an avatar inside a virtual dance studio. Users interact with a camera and tripod to film content, a makeup station to browse and apply filters, and a dance mirror to learn and gain mastery of dances. This prototype tests two of the three core interactions: filming and filter selection.

Testing Objective

From my design concept report, I had the assumption that TikTok-derived interface conventions, such as button presses, swipe-style browsing, grab-and-place interactions, were easily discoverable, usable, and intuitive. However, in practice, these assumptions are unvalidated and unfounded in XR.

Therefore, this test aims to discover whether first-time users can complete three core tasks: starting/stopping recording; placing the camera on a tripod; browsing/selecting a filter. The test would be conducted with minimal instruction (besides which commands to use to view and interact with XR), how long each task takes, and where users hesitate or make errors.

Testing Methodologies

This testing plan uses multiple different tests to evaluate the use of the digital prototype in Unity, such as task-based usability test, think-aloud protocol, and task-completion, to evaluate a digital prototype made in Unity. Participants will be given minimal instruction, asked to verbalise their thoughts as they attempt each task, and observed for completion time, success/failure, and points of confusion throughout. These data results will then be logged and submitted to GitHub at the end of the class.

Prototype description/requirements

The prototype was designed to let a participant attempt three core interactions inside a VR dance studio environment, allowing me to observe discoverability and ease of use without guidance. These include: a grabbable camera prop with a Record/Stop toggle button; a tripod that can be grabbed and positioned; and a mirror makeup station with a browsable filter carousel (left/right arrow buttons to cycle filters, a Confirm button to select, and an Exit button to close the interface). The environment also includes a dance mirror for users to learn dances, though this interaction has not been implemented yet.

Data collection method

During the testing process, I will be timing each of the 3 tasks from start to completion, noting success/failure of the task, any errors or hesitations, and recording think-aloud comments and questions where relevant. After all 3 tasks, I will ask participants for a 1–5 ease-of-use rating per task and a short verbal reflection on what felt intuitive or confusing.

Testing Setup

- Have the Unity prototype built and running on a laptop before each participant arrives.
- Prepare a simple paper log sheet with columns for: participant number, task, time taken, success/fail, errors observed, think-aloud notes, ease-of-use rating, and additional comments made throughout their testing period
- Have a stopwatch/timer ready on the phone.
- Brief participants with a short, neutral script (see Testing process below) so every participant gets the same instructions. The script is intentionally simple so that I can see which questions they ask, and which interactions are intuitive.

Testing process: (also considering the schedule/time)

1. Welcome the participant, explain they'll try 3 short tasks in VR. I will explain each task that I want them to do, and the testing method I would like to implement. There are no wrong answers and no bad questions, as I am testing the design, not their competition of the task
2. *Task 1 - Recording*: "Find a way to start recording with the camera, then stop it." Start timer. Observe and log time, success/fail, errors.
3. *Task 2 - Tripod*: "Now place the camera onto the tripod. Please verbalise your actions out loud when doing this." Start timer. Observe and log time, success/fail, errors, and participant's verbalisation.
4. *Task 3 - Filters*: "Go to the mirror and browse through the filters, select one, then exit." Start timer. Observe and log time, success/fail, errors.Ask the participant to rate each of the 3 tasks 1–5 for ease of use, and ask: "What felt intuitive? What was confusing?"
5. Thank the participant and log final notes before the next person begins.